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Operative vs Nonoperative Management of Fractures of the Humeral Diaphysis
The Humeral Shaft Fracture Fixation Randomized Clinical Trial

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IMPORTANCE Humeral shaft fractures are routinely managed nonoperatively, but this approach is potentially associated with higher nonunion rates and inferior functional outcomes when compared with operative fixation.

OBJECTIVE To assess whether there is any difference in outcome between surgery and functional bracing for adults with an isolated, closed humeral shaft fracture.

DESIGN, SETTING, AND PARTICIPANTS This prospective, superiority, parallel-group randomized clinical trial was conducted between September 2018 and October 2023 and took place at an academic major trauma center in the United Kingdom. Patients were reviewed at 2 and 6 weeks and 3, 6, and 12 months postintervention. Patients included 70 adults with an isolated, closed humeral shaft fracture. Exclusion criteria included absolute indications for surgery, pathological/periprosthetic fractures, multiple traumas, significant frailty, and inability to comply with follow-up. Data were analyzed from November 2023 through January 2024.

INTERVENTIONS Open reduction and plate fixation (n = 36) or functional bracing (n = 34). Seven patients did not receive their assigned treatment (operative, 5; nonoperative, 2).

MAIN OUTCOMES AND MEASURES The primary outcome measure was the Disabilities of the Arm, Shoulder, and Hand score (DASH) at 3 months postintervention. Secondary outcomes included health-related quality of life (EuroQol 5-Dimension [EQ-5D]/health visual analog scale [EQ-VAS] and Short Form [SF]-12 Physical Component Summary [PCS]/Mental Component Summary [MCS] scores), pain, shoulder/elbow range of motion, and complications. Intention-to-treat analyses were used.

RESULTS The study included 70 patients (mean [SD] age, 49 [17.1] years; 38 female [54%] and 32 male [46%]). At 3 months, 66 patients (94%) had completed follow-up. The operative group had a significantly better DASH score (difference, 15.0; *P* = .01). Surgery was also associated with a superior DASH score at 6 weeks (difference, 14.7; *P* = .01), but not at 6 months (*P* = .10) or at 12 months (*P* = .78). Surgery was further associated with a higher EQ-5D score (6 weeks: difference, 0.126, *P* = .03), EQ-VAS score (6 months: difference, 7; *P* = .04), and SF-12 MCS score (6 weeks: difference, 9.3; *P* = .001; 3 months: difference, 6.9; *P* = .01; and 6 months: difference, 7.1; *P* = .01). Brace-related dermatitis was significantly more common in the nonoperative group (18% vs operative 3%; *P* = .05). There were 8 nonunions (11%; operative 6% vs nonoperative 18%, *P* = .14).

CONCLUSIONS AND RELEVANCE For patients with a humeral shaft fracture in this study, surgery conferred early functional advantages over bracing. However, these benefits should be considered in the context of potential operative risks and the absence of any difference in outcomes at 1 year.

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Visual Abstract

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Humeral shaft fractures (HSFs) comprise 1% of all adult fractures,[1] with an annual incidence of 12 per 100 000.[2] Although specific indications exist for initial operative fixation, there is uncertainty regarding the optimal management of patients with isolated, closed HSFs. Functional bracing is the preferred strategy in many centers.[3] However, the increased rate of nonunion following bracing[4] and the risk of inferior longer-term patient-reported outcomes (PROs), even when union is achieved following nonunion surgery,[5,6] may support broadening the role of primary fixation.

Several recent meta-analyses have suggested potential advantages of surgery over bracing in terms of malunion, nonunion, reoperation,[7-9] and functional outcomes.[8] However, all have included nonrandomized or retrospective studies. There is a paucity of level 1 evidence comparing operative and nonoperative management,[4] with only 4 published randomized clinical trials (RCTs) currently available (comprising 292 patients in total).[10-13] Furthermore, only 1 RCT[13] has compared the operative standard of open reduction and plate fixation with functional bracing; this trial involved a relatively small sample size, with a high rate of crossover from bracing to surgery (29% at 12 months).[13] High-quality RCT data comparing surgery with bracing are essential to inform the management of patients with HSFs.

The primary aim was to determine whether any functional difference exists between operative (surgical fixation) and nonoperative management (functional bracing) for patients with an HSF. The null hypothesis was that there was no difference in limb-specific functional outcome between these treatment options. The secondary aim was to determine whether any difference exists in other clinically important outcomes (patient-reported outcome measures, shoulder/elbow range of motion, treatment complications, fracture union, return to work and sport) between surgery and bracing in the year following intervention.

Key Points

Question Is there any difference in functional outcome between surgery and bracing for humeral shaft fractures?

Findings This randomized clinical trial of 70 patients found surgery conferred superior upper limb function at 6 weeks and at 3 months, but not at 6 months or at 12 months. Surgery was associated with superior early shoulder/elbow motion, pain scores, and health-related quality of life; there was an increased risk of dermatitis with bracing, but no differences in other complications.

Meaning In this study, surgery conferred early functional advantages over bracing for humeral shaft fractures; these should be considered in the context of operative risks and the absence of any difference in outcomes at 12 months.

Methods

The HU-FIX Study was a superiority design, pragmatic, prospective RCT that followed the Consolidated Standards of Reporting Trials (CONSORT) reporting guidelines.[14] The study was performed at a single, tertiary-referral academic major trauma center in the United Kingdom between September 2018 and October 2023. The study received ethical approval in July 2018 (18/SS/0073) and was registered prospectively (NCT03689335). The study protocol was published prior to recruitment completion.[15]

Participants

All adults (16 years or older) with an isolated, closed fracture of the humeral diaphysis were considered for inclusion. Exclusion criteria were absolute indications for nonoperative (completely undisplaced fracture) or operative management (open fractures, progressive neurological deficit), pathological/periprosthetic fractures, additional spine/limb injuries (affecting rehabilitation), patients medically unfit for surgery, a Clinical Frailty Scale score of 6 or more,[16] or inability to comply with follow-up. Recruitment and flow of participants is shown in **Figure 1**. A temporary recruitment suspension, due to the COVID-19 pandemic, meant 10 potential participants were not approached.

Randomization

Allocation followed a parallel assignment model, with a 1:1 ratio of allocation into each treatment arm. Using block randomization with a mixed block size, a computer-generated randomization schedule (stratified by age; younger than 65 years and 65 years or older) was produced by an independent statistician using nQuery Advisor version 7.0 (Statsols). Research administrators, independent of study recruitment, used this to produce a series of sequentially numbered opaque sealed envelopes that indicated treatment allocation.

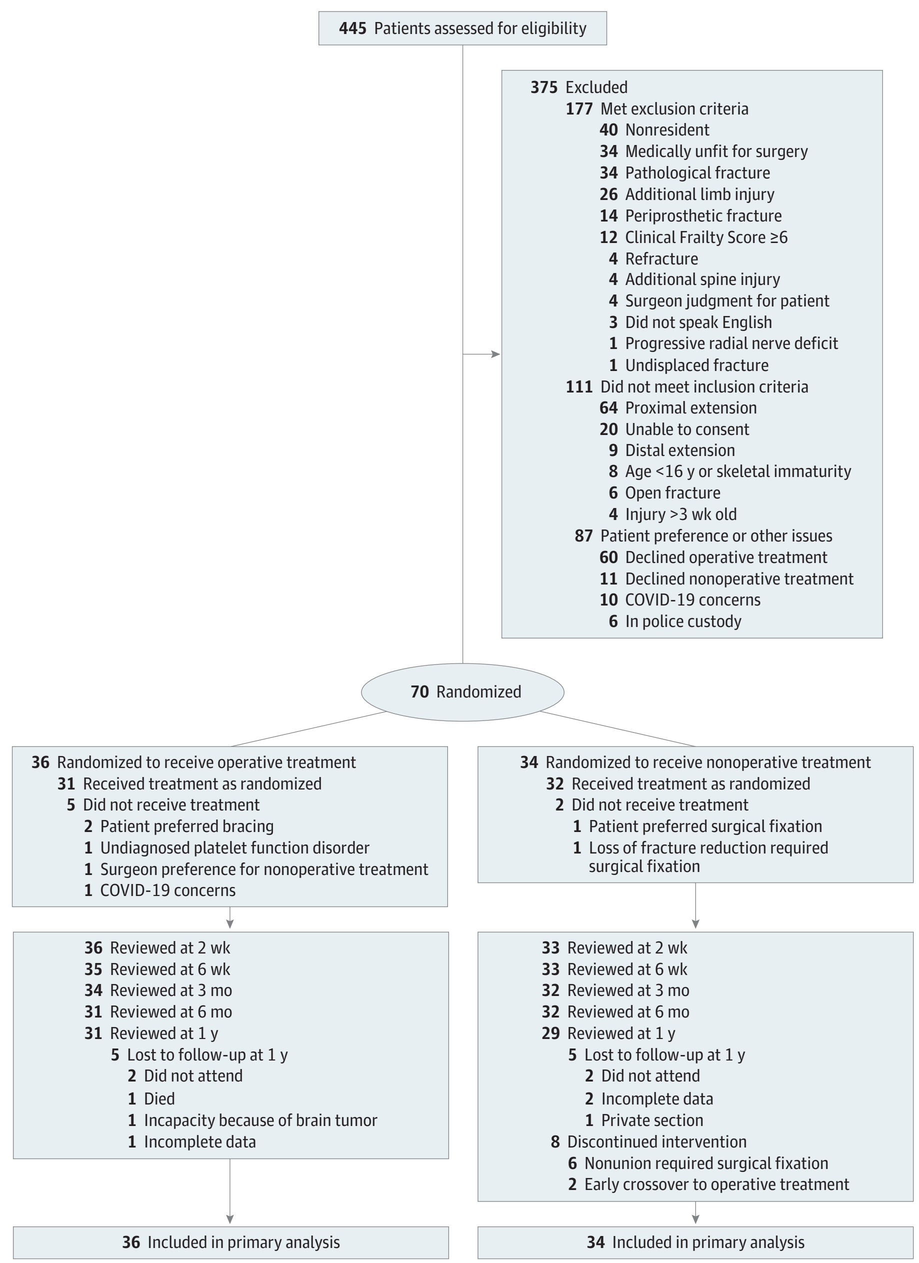
Interventions

All patients presented to the emergency department and were initially immobilized in either a plaster of Paris U-slab (42 of 70 [60%]), above-elbow cast (7 of 70 [10%]), collar-and-cuff sling (3 of 70 [4%]), or functional brace (18 of 70 [26%]), based on clinical assessment and fracture configuration.

Definitive care was overseen by 1 of 12 consultant orthopedic trauma surgeons. Patients allocated to operative management underwent surgery, using a standard technique of open reduction and plate-and-screw fixation. All other aspects, including surgical approach, fixation technique (compression/bridge plating, use of interfragmentary lag screws), wound closure, postoperative immobilization, and range of motion (ROM) restrictions, were at the treating surgeon's discretion consistent with the pragmatic design.

Patients allocated to nonoperative management were fitted with a functional brace in the clinic within 2 weeks of injury. Two humeral brace models were used during the study period—the Clasby brace (Beagle Orthopaedic) and the ProCare over-the-shoulder brace (DJO Global). The standard of care was to use the Clasby brace for proximal or mid-shaft fractures and the Procare brace for distal/multifragmentary fractures. The brace was worn until clinical and radiographic union. Patients were instructed to begin passive pendular shoulder exercises with full elbow, wrist, and

Figure 1. CONSORT Flow Diagram



hand mobilization as soon as the brace was applied. Physiotherapy was arranged at the discretion of the treating surgeon; 64 patients (91%) received 1 or more physiotherapy session (median, 5; range, 0-24).

Outcome Assessment

Given the study interventions, it was not feasible to blind participants, surgeons, or assessors to treatment allocation. All outcome assessment was performed alongside clinic follow-up at 2 weeks, 6 weeks, 3 months, 6 months, and 12 months postintervention. At each visit, physical examination was performed, complications recorded, and outcome scores obtained using a patient-reported questionnaire. Preinjury patient-reported outcomes were obtained at the first clinic visit (within 2 weeks postinjury). During the COVID-19 pandemic, several protocol amendments were authorized to allow for remote follow-up of participants, including obtaining PROs over the telephone and provisions for participants to submit digital images of their shoulder and elbow ROM.

The primary outcome measure was the Disabilities of the Arm, Shoulder and Hand (DASH) score at 3 months postintervention (0 = no disability; 100 = worst disability).[17] The DASH score has been demonstrated to be valid and reliable in evaluating HSF outcomes.[18] The primary end point of 3 months postintervention was to assess whether one treatment method conferred earlier functional recovery compared to the other.

Table 1. Baseline Patient and Injury Characteristics

Characteristic	No. (%) Operative (n = 36)	Nonoperative (n = 34)
Sex		
Male	18 (50)	14 (41)
Female	18 (50)	20 (59)
Age at injury, y, mean (SD) [95% CI]	49.7 (17.2) [43.9-55.5]	48.8 (18.5) [42.4-55.3]
Age at injury, y		
<65	27 (75)	26 (76)
≥65	9 (25)	8 (24)
Comorbidities		
0	19 (53)	14 (41)
≥1	17 (47)	20 (59)
BMI,[a] median (IQR)	26.2 (23.4-30.8)	28.2 (24.0-33.2)
Current smoker		
No	23 (64)	25 (74.5)
Yes	13 (36)	9 (26.5)
Alcohol consumption, units/wk, median (IQR)	12 (0-20)	4 (0-12)
SIMD quintile		
1 (Most deprived)	4 (11)	6 (18)
2	8 (22)	6 (18)
3	6 (17)	4 (12)
4	7 (19)	13 (38)
5 (Least deprived)	11 (31)	5 (15)
Clinical Frailty Scale		
1 (Very fit)	20 (56)	13 (38)
2 (Fit)	8 (22)	12 (35)
3 (Managing well)	7 (19)	7 (21)
4 (Very mild frailty)	1 (3)	2 (6)
Employment status		
Employed	23 (64)	23 (68)
Unemployed/retired	13 (36)	11 (32)
Sports participation		
Yes	20 (56)	15 (44)
No	16 (44)	19 (56)
Injury mechanism		
Fall from standing	22 (61)	18 (53)
Fall from height	2 (6)	4 (12)
Direct blow	3 (8)	3 (9)
Road traffic collision	5 (14)	3 (9)
Sport	2 (6)	2 (6)
Other	2 (6)	4 (12)
Side of injury		
Right	16 (44)	20 (59)
Left	20 (56)	14 (41)
Dominant	16 (44)	17 (50)
Nondominant	20 (56)	17 (50)
Fracture location		
Proximal	13 (36)	10 (29)
Middle	15 (42)	16 (47)
Distal	8 (22)	8 (24)
AO-OTA type		
A	25 (69)	26 (77)
B	10 (28)	7 (21)
C	1 (3)	1 (3)
Radial nerve palsy		
No	33 (92)	31 (91)
Yes	3 (8)	3 (9)

Abbreviations: AO-OTA, Arbeitsgemeinschaft für Osteosynthesefragen-Orthopaedic Trauma Association; BMI, body mass index; SIMD, Scottish Index of Multiple Deprivation.

[a] Calculated as weight in kilograms divided by height in meters squared.

Secondary PROs included EuroQol (EQ) scores, including the 5-dimension 3-level health index (EQ-5D)[19] and health visual analog scale (EQ-VAS: 0 = worst health; 100 = perfect health); 12-item Short Form (SF-12) health survey; Physical Component Summary (PCS) and Mental Component Summary (MCS) scores[20]; satisfaction VAS, relating to treatment and arm appearance (0 = completely dissatisfied, 10 = completely satisfied); pain VAS in the affected arm (0 = no pain; 10 = worst pain), or body in general (0 = no pain; 100 = worst pain). Other secondary outcomes included shoulder (elevation, abduction, external rotation, internal rotation) and elbow ROM (flexion, extension), treatment complications (neurovascular injury, superficial/deep infection, metalwork prominence, fixation failure, periprosthetic fracture, nonunion, revision/further surgery, medical complications including death), and radiographic alignment (using standard anteroposterior and lateral radiographs). ROM was measured in degrees using a standard full-circle goniometer and presented as a percentage of the contralateral side (lower percentage = worse ROM). Internal rotation was measured as the distance from the C7 spinous process to the tip of the thumb (in centimeters),[21,22] and presented as the deficit compared with the contralateral side (in centimeters; larger distance = worse ROM). Complications were defined as either minor (if not requiring further surgery) or major (if requiring further surgery). The rate and timing of return to work and sport were recorded.

Statistical Analysis

Based on a DASH score minimal clinically important difference of 10 points[23,24] and SD of 12 points,[25] an a priori power calculation determined 64 participants (32 in each group) were required to generate 90% power to detect a meaningful difference between groups. Assuming approximately 10% loss-to-follow-up, we planned to recruit 70 participants to the study (35 in each group). Intention-to-treat (ITT) analyses were performed with supplementary per protocol (PP) and as-treated (AT) analyses where appropriate.

Continuous data were compared using a 2-sided, independent-samples *t* test or Mann-Whitney *U* test, as indicated

Table 2. Patient-Reported Outcome Measures

Outcome	Median (IQR) Operative	Median (IQR) Nonoperative	P value
Preinjury (n = 70)			
DASH,	0 (0-0.8)	0 (0-1.7)	.86
EQ-5D	1 (1-1)	1 (1-1)	.39
EQ-VAS	92 (80-100)	90 (80-100)	.15
SF-12 PCS	56.2 (54.9-56.6)	55.7 (49.1-57.1)	.47
SF-12 MCS	57.8 (53.7-60.7)	55.4 (46.0-58.8)	.09
6 wk (n = 68)			
DASH, mean (SD) [95% CI]	38.4 (21.0) [31.2-45.6]	53.1 (20.7) (45.8-60.5)	.01[a]
EQ-5D	0.760 (0.516-0.760)	0.585 (0.258-0.689)	.03[a]
EQ-VAS	80 (80-90)	70 (60-90)	.15
SF-12 PCS, mean (SD) [95% CI]	37.8 (7.3) [35.3-40.3]	34.7 (6.4) [32.5-37.0]	.07
SF-12 MCS	54.7 (44.2-60.8)	42.5 (32.4-56.1)	.001[a]
3 mo (n = 64)			
DASH, mean (SD) [95% CI]	24.5 (20.4) [17.3-31.6]	39.4 (22.6) [31.3-47.6]	.01[a]
EQ-5D	0.796 (0.689-0.913)	0.673 (0.533-0.861)	.07
EQ-VAS	90 (70-95)	80 (60-90)	.01
SF-12 PCS	47.2 (36.6-52.6)	38.4 (33.7-48.6)	.06
SF-12 MCS	57.8 (49.0-60.3)	48.0 (35.7-57.4)	.01[a]
6 mo (n = 63)			
DASH	6.9 (1.7-19.0)	14.2 (2.8-46.4)	.01
EQ-5D	0.796 (0.760-1)	0.796 (0.585-1)	.34
EQ-VAS	90 (80-100)	80 (70-93)	.04[a]
SF-12 PCS	50.5 (44.7-55.5)	47.4 (34.6-55.5)	.33
SF-12 MCS	57.9 (54.4-60.8)	53.2 (41.9-57.8)	.01[a]
12 mo (n = 60)			
DASH	5.0 (1.7-12.5)	4.3 (0.8-24.6)	.78
EQ-5D	0.796 (0.760-1)	0.796 (0.672-1)	.58
EQ-VAS	90 (80-100)	90 (70-98)	.50
SF-12 PCS	54.6 (45.8-56.6)	52.0 (39.6-56.6)	.34
SF-12 MCS	54.1 (46.2-60.7)	54.1 (44.0-57.0)	.20

Abbreviations: DASH, Disabilities of the Arm, Shoulder and Hand score; EQ-5D, EuroQol 5-Dimension 3-Level health index; EQ-VAS, EuroQol Health Visual Analogue Scale; SF-12 MCS, 12-item Short-Form Mental Component Summary score; SF-12 PCS, 12-item Short-Form Physical Component Summary score.

[a] $P < .05$.

by data normality. The change in DASH score during study follow-up was compared using 2-way repeated-measures analysis of variance. Binary outcomes were compared using a binomial test for the comparison of proportions. Missing data were not imputed. Significance was set at $P < .05$; 95% CIs and 2-tailed P values were reported.

Results

Seventy patients (mean [SD] age, 49 [17.1] years [95% CI, 45.1-53.5; range, 18-81 years]; 38 female [54%] and 32 male [46%]) were randomized to either surgery (36 of 70) or bracing (34 of 70). There were no baseline differences between the groups (**Table 1**). Six patients (9%; 3 per group) had a radial nerve palsy at presentation, all of which were managed expectantly in a wrist splint and resolved at a median of 9 weeks (IQR, 5.0-32.5 weeks).

Of 36 patients randomized to surgery, 31 underwent fixation and 5 did not receive their allocation and were managed nonoperatively (Figure 1). Perioperative details are available in eTable 1 in Supplement 1. Of 34 patients randomized to bracing, 2 underwent surgery after randomization and 1 was lost to follow-up prior to completing their treatment (Figure 1). The remaining patients were managed in a functional brace; the median duration of immobilization was 87 (IQR, 82-97) days.

Primary Outcome

Sixty-six patients (94%) completed 3-month follow-up (primary end point). The mean score for the operative group (24.45) was lower than the nonoperative group (39.44) with a difference of 14.99 points (SD, 21.51; 95% CI, 4.41-25.58; $P = .01$).

Surgery was associated with a superior DASH score at 6 weeks, but no difference at 6 or 12 months (**Table 2**; **Figure 2**). The change in the DASH score over the course of 1-year follow-up was superior in the operative group (marginal mean difference, 11.51 points; 95% CI, 3.17-19.85; $P = .01$).

Secondary Outcomes

Health-Related Quality of Life

The operative group reported superior health-related quality of life (HRQoL) according to the EQ-5D at 6 weeks (difference, 0.175; $P = .03$) and EQ-VAS at 6 months (difference, 10; $P = .04$;

Table 2). The operative group also reported superior SF-12 MCS at 6 weeks (difference, 12.2; *P* = .001), 3 months (difference, 9.8; *P* = .01), and 6 months (difference, 4.7; *P* = .01; Table 2). There were no differences in HRQoL at 12 months.

Satisfaction

At 6 months, the operative group were more satisfied with their treatment (median 100/100; IQR, 99-100 vs median 100/100; IQR 83-100; *P* = .05). There were no other differences in satisfaction with treatment or arm appearance (eTable 2 in Supplement 1).

Pain

The operative group reported better (lower) arm pain scores at 2 weeks (difference, 1.8/10; *P* = .01) and 6 months (difference, 1/10; *P* = .03; eTable 3 in Supplement 1). The operative group also reported better (lower) overall body pain scores at 6 weeks (difference, 20/100; *P* = .03) and 6 months (difference, 10/100; *P* = .02). There were no differences in arm or body pain scores at 12 months.

Range of Motion

The operative group had a superior recovery of shoulder elevation, abduction, and external rotation at 6 weeks and 3 months (eTable 4 in Supplement 1). There was also statistically superior recovery of elbow flexion at 3 months (difference, 6%; *P* = .001) and 12 months (difference, 3%; *P* = .05). There were no other differences in ROM at 6 months or 12 months.

Complications

Treatment-related complications occurred in 35 of 69 patients (51%) (**Table 3**). In the operative group, 18 of 36 (50%) developed 1 or more complications (minor, 14 [39%]; major, 4 [11%]), compared with 17 of 33 (52%) in the nonoperative group (minor, 11 [33%]; major, 7 [21%]). There was no difference in the rate of minor, major, or overall complications

Eight patients (11%) developed a nonunion; this was more common after bracing (18% vs 6%) but did not reach statistical significance (*P* = .14). Six patients required nonunion surgery at a median of 25 weeks postinjury (range, 12-42 weeks). One patient, who did not undergo their allocated surgery due to a previously undiagnosed platelet function disorder, developed a nonunion but was asymptomatic and opted for continued nonoperative management. One patient in the operative group developed a nonunion following plate fixation, but was similarly asymptomatic and did not undergo reoperation.

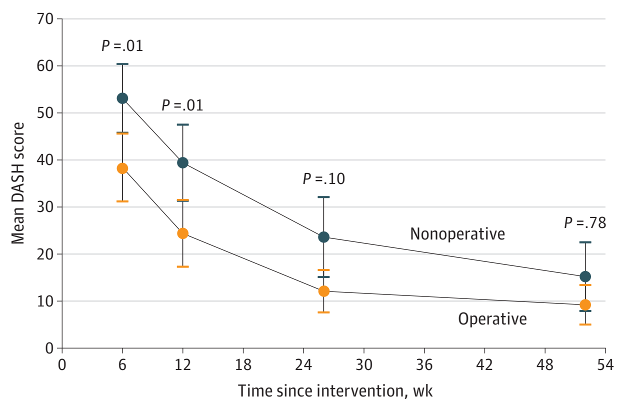
Three patients (9%) developed a transient-postoperative radial nerve palsy. All were successfully managed in a wrist splint and resolved at a median of 6 (IQR, 3-6) months. No cases of permanent radial nerve palsy were observed.

Brace-related dermatitis was more common in the nonoperative group (18% vs 3%; *P* = .05), but all cases were successfully managed with dressings and/or emollient creams. Three patients managed nonoperatively also developed superficial infection, requiring either topical (n = 1) or oral antibiotics (n = 2). One patient in the operative group developed a superficial wound infection (*Staphylococcus aureus*), resulting in partial wound dehiscence. This was treated with dressings and extended course oral antibiotics; following fracture union, the patient opted to undergo metalwork removal.

One patient, initially allocated to nonoperative management but who underwent surgery due to early loss of reduction, developed a deep postoperative infection (*Enterobacter cloacae*). They underwent successful open debridement and washout 8 days postoperatively, followed by a 3-month course of antibiotic therapy. Two patients (6%) had symptomatic metalwork prominence postoperatively, with 1 requiring implant removal. There was 1 case of adhesive shoulder capsulitis in the operative group that partially resolved following distension arthrography and 1 case of long head of biceps tendinitis managed with ultrasound-guided injection.

There was 1 death in the operative group. This was an elderly patient who, following initial fixation, underwent multiple subsequent surgeries and received intravenous antibiotics for fixation failure and a polymicrobial deep infection (*Staphylococcus epidermidis* and *Pseudomonas aeruginosa*), ultimately requiring a fasciocutaneous flap reconstruction. This flap reconstruction failed and the patient sustained a further periprosthetic fracture of the distal humerus. While undergoing surgery for removal of metalwork and wound debridement, the patient had a fatal intraoperative stroke.

Figure 2. Disabilities of the Arm, Shoulder, and Hand Scores (Intention-to-Treat Analysis)



Radiographic Alignment

Three patients (5%) developed a malunion (2 varus; 1 procurvatum), all occurring after bracing, but the difference did not reach significance. There was an increased rate of healing in varus in the nonoperative group (80% vs 32%; *P* < .001), with increased angulation in the coronal (median, 8° *vs* 2°; *P* < .001) and sagittal planes (median, 6.5° vs 2°; *P* < .001; eTable 5 in Supplement 1).

Return to Activity

Of 47 patients in employment, follow-up was available for 40 (85%). Thirty-three returned to work (83%) at a median of 8.6 (IQR, 2-14) weeks. The rate (operative, 16 of 19 [84%] vs nonoperative, 17 of 21 [81%]; *P* = 1.00) and median time of return

Table 3. Management-Related Complications

Complication type	No. (%) Operative (n = 36)	Nonoperative (n = 33)	*P* value
Minor complications			
Transient radial nerve palsy	3 (8)	0	.24
Dermatitis	1 (3)	6 (18)	.05[a]
Superficial infection	1 (3)	3 (9)	.34
Wound breakdown/dehiscence	1 (3)	0	1.00
Prominent metalwork (no removal)	1 (3)	0	1.00
Ultrasound-guided shoulder injection	2 (6)	0	.49
Major complications			
Loss of reduction	0	1 (3)	.48
Deep infection	2 (6)	1 (3)	1.00
Failure of fixation	1 (3)	0	1.00
Periprosthetic fracture	2 (6)	0	.49
Nonunion	2 (6)	6 (18)	.14
Prominent metalwork (removal)	1 (3)	0	1.00
Secondary surgery	3 (8)	7 (21)	.18
Death	1 (3)	0	1.00

[a] $P < .05$.

to work (operative, 7.3 weeks vs nonoperative, 10 weeks; $P = .75$) were not statistically different.

Of 35 patients playing sport prior to injury, follow-up was available for 32 (91%). Twenty-five returned to sport (78%) at a median of 13 (IQR, 10-26) weeks. The rate (17 of 18 [94%] vs nonoperative, 8 of 14 [57%]; $P = .03$) of return was superior in the operative group, but the median time of return (12 weeks vs 16.3 weeks; $P = .17$) was not.

Other Analyses

Given the crossover rate (10%), DASH scores (eTable 6, eFigure 1, eTable 7, and eFigure 2 in Supplement 1) and complications (eTable 8 and eTable 9 in Supplement 1) were also summarized based on PP and AT analyses. There were no differences in the findings between these analyses and the ITT analysis.

Discussion

This RCT demonstrated that surgery conferred superior early upper-limb function compared with bracing for patients with an isolated, closed HSF. Further advantages of fixation included superior early HRQoL and pain scores, superior recovery of upper limb ROM, and an increased rate of return to sport. These early benefits present a compelling case for offering surgery to low-risk patients, but should be weighed against the potential for operative complications, particularly in higher-risk groups. Given the absence of any differences in PROs at 1 year, it is uncertain whether a strategy of routine fixation for all patients with HSFs is indicated.

Consistent with previous RCTs comparing operative and nonoperative management,[10,11,13] surgery conferred superior patient-reported upper-limb function up to 3 months postoperatively. The current study also demonstrated superior HRQoL in favor of surgery, that appeared to continue for between 3 months (EQ-5D) and 6 months (EQ-VAS) postintervention. The surgical group reported a superior SF-12 MCS up to 6 months, which may reflect the psychological benefits of early fixation from the patient's perspective. Matsunaga et al[11] found a difference in the physical functioning domain of the SF-36 at 1 month, with no difference in any other domains up to 1 year. Rämö et al[5,13] found no differences in HRQoL up to 2 years. These data suggest that surgery may confer early benefits in terms of upper limb function and HRQoL, but these do not persist beyond 6 months.

To our knowledge, this is the first RCT to specifically document upper-limb ROM after HSF and surgery was associated with superior recovery up to 3 months. This, along with improved early arm and body pain scores, may partly underpin the superior early DASH in the operative group, but clinically important differences in ROM are unknown and the influence of small differences upon outcome is uncertain.[26] Furthermore, previous RCTs have reported no differences between surgery and bracing at 6 months in terms of pain on activity or at rest,[11,13] with pooled analysis of this data appearing to favor nonoperative management.[4] The reason for better pain scores among patients managed surgically in this study is unclear, although the effect size was small and there was no difference in pain scores at 1 year.

In accordance with some previous data[11,13] the current findings indicate the initial functional benefits of surgery appear to be transitory, with no differences in patient-reported function, HRQoL, satisfaction, pain, ROM, or return to employment at 1 year. Given this, it could be argued that functional bracing should remain the preferred strategy for most patients. One advantage of surgery is avoiding brace-related complications, including dermatitis, loss of fracture reduction, and the need for delayed surgical intervention, particularly for nonunion. Dermatitis affected 1 in 5 patients managed nonoperatively in this study, with a further 8% developing superficial infection. Matsunaga et al[11] reported dermatitis in 11% of their bracing group. In the present study, 1 patient sustained a loss of reduction in their brace necessitating early surgical intervention. Matsunaga et al[11] reported 2 patients (4%) in the bracing group were intolerant of their treatment and underwent fixation. Rämö et al[13] identified 4 patients who crossed over to surgery within 6 weeks due to excess pain, intolerance of bracing, or loss of reduction. One patient in the latter study also developed a 35° varus malunion following bracing, complicated by a refracture at 8 months postinjury. The increased likelihood of secondary surgery following initial bracing is recognized.[4]

A meta-analysis of level 1 evidence found a substantial difference in pooled nonunion rate between surgery and bracing[4]; this was also apparent in the present study, but did not reach significance. A larger study would be needed to detect a significant difference, although a 3-fold higher rate of nonunion in the nonoperative group is likely to be clinically relevant. A lower risk of nonunion is an important consideration when counselling patients about the potential advantages of surgery, particularly given the inferior outcome of successful nonunion surgery when compared to primary union.[5,6]

The rate of transient radial nerve palsy and superficial infection in this study are consistent with previous RCTs.[10-13] This trial also highlighted the potential for devastating operative complications, with 2 patients developing a deep postoperative infection, 1 of whom died following further surgery. Although bracing does have limitations, it confers less risk of serious adverse events, particularly in older patients with comorbidities for whom the risks of anesthesia and major surgery should always be carefully considered.

Limitations

The study was performed in a single center that may limit generalizability of the findings, although the center provides orthopedic trauma care for a large catchment population.[1,2] Moreover, the study cohort likely reflects typical demographics of patients with HSF more closely than existing RCTs.[2] Blinding of patients and surgeons was not possible given the interventions being compared and assessor blinding was not feasible as the presence/absence of a surgical wound/scar and metalwork on radiographs were obvious during outcome assessment. Preinjury patient-reported outcomes were obtained as soon as possible after injury; although common to all trauma research studies, this nonetheless introduces potential for recall bias. Although the DASH has been widely used in previous RCTs,[10,11,13] it does not distinguish between injured and uninjured limbs, which may inflate scores for patients better able to compensate for loss of function in 1 limb. We also recognize there is limited data relating to the DASH SD and minimal clinically important difference in the context of fracture.[18] It is acknowledged that only 90% of participants (63 of 70) received their initial treatment allocation and the rate of crossover may have limited our ITT analysis. However, this is accounted for in our AT and PP analyses. The study was likely underpowered to detect a difference in the DASH at 6 months and 12 months, due to loss to follow-up. Lastly, there may be subgroups of patients, according to age and preinjury level of function, that would benefit from fixation, which was not assessed in the current study.

Conclusions

In conclusion, surgery for an isolated closed HSF conferred early advantages in terms of upper-limb function, HRQoL, pain, and ROM. However, there were no differences in PROs beyond 6 months, and thus, these transient functional benefits should be weighed carefully against the risk of operative complications. Early surgery targeting patients at the greatest risk of nonunion may represent the most appropriate strategy.

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REFERENCES

1. Court-Brown CM, Caesar B. Epidemiology of adult fractures: a review. *Injury*. 2006;37(8):691-697. doi:10.1016/j.injury.2006.04.130

2. Oliver WM, Searle HKC, Ng ZH, et al. Fractures of the proximal- and middle-thirds of the humeral shaft should be considered as fragility fractures. *Bone Joint J*. 2020;102-B(11):1475-1483. doi:10.1302/0301-620X.102B11.BJJ-2020-0993.R1

3. Sarmiento A, Kinman PB, Galvin EG, Schmitt RH, Phillips JG. Functional bracing of fractures of the shaft of the humerus. *J Bone Joint Surg Am*. 1977;59

(5):596-601. doi:10.2106/00004623-197759050-00004

4. Oliver WM, Bell KR, Molyneux SG, White TO, Clement ND, Duckworth AD. Surgical versus nonsurgical management of humeral shaft fractures: a systematic review and meta-analysis of randomized trials. *J Am Acad Orthop Surg*. 2023;31(2):e82-e93. doi:10.5435/JAAOS-D-22-00432

5. Rämö L, Paavola M, Sumrein BO, et al; FISH Investigators. Outcomes with surgery vs functional bracing for patients with closed, displaced humeral shaft fractures and the need for secondary surgery: a prespecified secondary analysis of the FISH randomized clinical trial. *JAMA Surg*. 2021;156(6):1-9. doi:10.1001/jamasurg.2021.0906

6. Oliver WM, Searle HKC, Molyneux SG, White TO, Clement ND, Duckworth AD. Factors associated with patient-reported outcomes following a humeral shaft fracture: nonunion results in a poorer outcome despite union after surgical fixation. *J Orthop Trauma*. 2022;36(6):e227-e235. doi:10.1097/BOT.0000000000002315

7. Beyer J, Rao B, Liu J, Skie M. Evaluation of humeral shaft fracture outcomes by treatment method: a systematic review and meta-analysis based on comparison studies. *JBJS Rev*. 2023;11(7). doi:10.2106/JBJS.RVW.23.00037

8. Van Bergen SH, Mahabier KC, Van Lieshout EMM, et al. Humeral shaft fracture: systematic review of non-operative and operative treatment. *Arch Orthop Trauma Surg*. 2023;143(8):5035-5054. doi:10.1007/s00402-023-04836-8

9. Zavras AG, Monahan KT, Winek NC, et al. Conservative management with functional brace versus various surgical fixation techniques for humeral shaft fractures: a network meta-analysis. *J Bone Joint Surg Am*. 2023;105(14):1112-1122. doi:10.2106/JBJS.22.01374

10. Kumar S, Shanmugam N, Kumar S, Ramanusan K. Comparison between operative and non operative treatment of fracture shaft of humerus: an outcome analysis. *Int J Res Orthop*. 2017;3(3):445-450. doi:10.18203/issn.2455-4510.IntJResOrthop20171537

11. Matsunaga FT, Tamaoki MJS, Matsumoto MH, Netto NA, Faloppa F, Belloti JC. Minimally invasive osteosynthesis with a bridge plate versus a functional brace for humeral shaft fractures: a randomized controlled trial. *J Bone Joint Surg Am*. 2017;99(7):583-592. doi:10.2106/JBJS.16.00628

12. Hosseini Khameneh SM, Abbasian M, Abrishamkarzadeh H, et al. Humeral shaft fracture: a randomized controlled trial of nonoperative versus operative management (plate fixation). *Orthop Res Rev*. 2019;11:141-147. doi:10.2147/ORR.S212998

13. Rämö L, Sumrein BO, Lepola V, et al; FISH Investigators. Effect of surgery vs functional bracing on functional outcome among patients with closed displaced humeral shaft fractures: the FISH randomized clinical trial. *JAMA*. 2020;323(18):1792-1801. doi:10.1001/jama.2020.3182

14. Schulz KF, Altman DG, Moher D; CONSORT Group. CONSORT 2010 statement: updated guidelines for reporting parallel group randomised trials. *Trials*. 2010;11(1):32. doi:10.1186/1745-6215-11-32

15. Oliver WM, Carter TH, Graham C, et al. A prospective randomised controlled trial of operative versus non-operative management of fractures of the humeral diaphysis: the HUmeral Shaft Fracture FIXation (HU-FIX) Study protocol. *Trials*. 2019;20(1):475. doi:10.1186/s13063-019-3576-0

16. Rockwood K, Song X, MacKnight C, et al. A global clinical measure of fitness and frailty in elderly people. *CMAJ*. 2005;173(5):489-495. doi:10.1503/cmaj.050051

17. Hudak PL, Amadio PC, Bombardier C; The Upper Extremity Collaborative Group (UECG). Development of an upper extremity outcome measure: the DASH (disabilities of the arm, shoulder and hand) [corrected]. *Am J Ind Med*. 1996;29(6):602-608. doi:10.1002/(SICI)1097-0274(199606)29:6<602::AID-AJIM4>3.0.CO;2-L

18. Mahabier KC, Den Hartog D, Theyskens N, Verhofstad MHJ, Van Lieshout EMM; HUMMER Trial Investigators. Reliability, validity, responsiveness, and minimal important change of the disabilities of the arm, shoulder and hand and Constant-Murley scores in patients with a humeral shaft fracture. *J Shoulder Elbow Surg*. 2017;26(1):e1-e12. doi:10.1016/j.jse.2016.07.072

19. EuroQol Group. EuroQol–a new facility for the measurement of health-related quality of life. *Health Policy*. 1990;16(3):199-208. doi:10.1016/0168-8510(90)90421-9

20. Ware J Jr, Kosinski M, Keller SDA. A 12-item short-form health survey: construction of scales and preliminary tests of reliability and validity. *Med Care*. 1996;34(3):220-233. doi:10.1097/00005650-199603000-00003

21. Han SH, Oh KS, Han KJ, Jo J, Lee DH. Accuracy of measuring tape and vertebral-level methods to determine shoulder internal rotation. *Clin Orthop Relat Res*. 2012;470(2):562-566. doi:10.1007/s11999-011-2112-7

22. Mitsukane M, Suzuki K, Tabe R, Hasumi F, Fukushima D. Normalized hand-behind-back for the measurement of shoulder internal rotation. *JSES Int*. 2022;6(2):287-291. doi:10.1016/j.jseint.2021.12.001

23. Roy JS, MacDermid JC, Woodhouse LJ. Measuring shoulder function: a systematic review of four questionnaires. *Arthritis Rheum*. 2009;61(5):623-632. doi:10.1002/art.24396

24. Franchignoni F, Vercelli S, Giordano A, Sartorio F, Bravini E, Ferriero G. Minimal clinically important difference of the disabilities of the arm, shoulder and hand outcome measure (DASH) and its shortened version (QuickDASH). *J Orthop Sports Phys Ther*. 2014;44(1):30-39. doi:10.2519/jospt.2014.4893

25. The DASH and QuickDASH Outcome Measure User's Manual (Third Edition). Accessed February 19, 2025. https://www.dash.iwh.on.ca/dash-manual

26. Jayakumar P, Overbeek CL, Lamb S, et al. What factors are associated with disability after upper extremity injuries? a systematic review. *Clin Orthop Relat Res*. 2018;476(11):2190-2215. doi:10.1097/CORR.0000000000000427

Invited Commentary

Options for Management of Fractures of the Humeral Diaphysis

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Humeral shaft fractures are common, yet the appropriate treatment remains elusive. There are absolute indications for surgery, such as those with open fractures that are prone to infection and those frail patients who should not have surgery. This clinical trial[1] recruited people who may fare well with either surgery or splinting. The authors found that at 12 months the outcomes were similar, but those who had fixation of the fractures tended to have better outcomes in the short term.

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The findings of this article[1] are not surprising, but it is a valiant effort to conduct a clinical trial to yield useful data for patients and surgeons to consider. When surgery is conducted expertly, patients will often recover quickly and be able to use the injured arm for activities. However, complications could occur, and these adverse events were not trivial. Because multiple surgeons performed the surgery, it is uncertain if all of them have the same experience and proficiency in treating this injury. A study on treating complex hand injuries showed that surgeon proficiency is highly influential in determining patient outcomes.[2] With nonoperative treatment, even if the humerus is malunited, the patients could have suitable outcomes, provided that the shoulder and elbow were functioning well to accommodate the deformity.

This study[1] is underpowered because the most important information that needs to be gleaned is what type of fracture, patient characteristics, and other modifiable factors were amenable to the treatment prescribed. For an active patient, splinting will not be an acceptable option because the patient will prefer to start activities as early as possible. An older patient with limited requirements and multiple comorbid conditions may opt for the nonoperative option.[3] These considerations require a subgroup analysis